

Two-shell 110+200 global no-condensate as a diagonal-continuum bound at the B1 point, with two operating-point corrections to the knob-closure note

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: the operating-point discrepancy

estimator-upgrade-knobs v1.0 certified the two-shell $\{110\} + \{200\}$ $(0,0)$ Hessian and cited Math432 for the GLOBAL no-condensate (the m_{31} tilted-valley question). Two problems surfaced on closer inspection. First, Math432 sets its bare mass $R = 0.005$ directly and reproduces $r_R = 0.3045$, $M_R = 0.1094$ – this is NOT the B1 operating point. The B1 single-shell readings use $r_{\text{bare}} = R_{\text{OF}}(0.005) = \mu^2 + q_0^4 = 0.219$, $r_R = 0.419$, $M_R = 0.096$. The cited global evidence is at a different (lower-mass) point. Second, the soft $(0,0)$ eigenvalue is NOT $\kappa_{\text{BCC}}(\{110\}) = 5.116$ but $\kappa(\{200\}) = 3.86$: the $\{200\}$ shell is the SOFTER direction. This note redoes the two-shell global no-condensate at the B1 point and corrects both.

2. The diagonal-continuum evaluator and its validation

The diagonal-continuum two-shell free energy $\Delta F_{\text{diag}}(A_1, A_2, M)$ (Hartree diagonal ansatz, relative to the disordered state) is built from the exact band-limited moments $\langle \phi_1^a \phi_2^b \rangle$ and the certified m_{424} primitives M_{fast} , **dI**. The moments are exact: $\langle \phi_1^2 \rangle = 12$, $\langle \phi_2^2 \rangle = 6$, $\langle \phi_1^4 \rangle = 540$, $\langle \phi_1^3 \phi_2 \rangle = 144$, $\langle \phi_1^2 \phi_2^2 \rangle = 96$, $\langle \phi_1 \phi_2 \rangle = \langle \phi_1 \phi_2^3 \rangle = 0$ (orthogonality). The evaluator reproduces Math432's recorded diagonal values (anchored minus bracket) at $r = 0.005$ to 1.6×10^{-7} over four anchors, so the re-implementation is faithful (the transcription is gated by these asserts).

3. Two-shell global no-condensate at the B1 point ($r = 0.219$)

Minimising ΔF_{diag} over the trial mass M (21-point grid, $M/M_R \in [0.5, 1.6]$) on the Math432 scan domain ($A_1 \in [0, 0.20]$, $A_2 \in [-0.20, 0.20]$, 51×51):

- **no diagonal condensate**: $\min_{(A_1, A_2) \neq (0,0)} \Delta F_{\text{diag}} = +3.9 \times 10^{-5} > 0$.
- **continuum (node-free)**: the 2D curvature-chord lower bound $\Delta F \geq \Delta F_{ij} - \frac{1}{8}(|D_{xx}| \delta_1^2 + |D_{yy}| \delta_2^2 + 2|D_{xy}| \delta_1 \delta_2)$ away from the origin cell is $+1.2 \times 10^{-4} > 0$: no condensate hides between grid nodes.
- **near origin**: the $(0,0)$ Hessian is positive-definite, $\kappa_{\{110\}} = 5.16$, $\kappa_{\{200\}} = 3.86$ (both > 0), $\kappa_{12} = 0$ by orthogonality.

Thus the two-shell diagonal global no-condensate holds at the B1 point, which supersedes the Math432 $r = 0.005$ citation.

4. Two corrections to estimator-upgrade-knobs v1.0

(1) **Operating point.** The two-shell GLOBAL no-condensate is now at $r = 0.219$ (B1), not the cited Math432 $r = 0.005$. The B1 point is STIFFER (larger gap $r_R = 0.419$ vs 0.305 ; larger $\{110\}$ curvature), so the no-condensate conclusion holds a fortiori, but it is now computed at the correct point.

(2) **Soft-direction identification.** estimator-upgrade-knobs v1.0 stated the soft eigenvalue is $\kappa_{\text{BCC}}(\{110\}) = 5.116$ and that $\{200\}$ is stiffer (kernel penalty $Cq_0^4 > 0$). The direct two-shell computation gives $\kappa_{\{200\}} = 3.86 < \kappa_{\{110\}} = 5.16$: $\{200\}$ is SOFTER. The penalty argument was naive – $\{200\}$ has fewer modes ($n_2 = 3 < n_1 = 6$) and the loop dressing nets a lower curvature. The $(0,0)$ Hessian is still positive-definite; only the soft-eigenvalue value (3.86, not 5.116) and the soft-direction label ($\{200\}$, not $\{110\}$) change.

5. What this advances and what remains

ADVANCED: the two-shell global no-condensate is upgraded from Math432 grid evidence at the wrong operating point to a DIAGONAL-continuum bound at the B1 point. REMAINING: the exact-Wick off-diagonal bracket ($\Delta F_{\text{anchored}} = \Delta F_{\text{diag}} + \text{bracket}$, bracket NEGATIVE, $|\text{bracket}| \leq 7.6 \times 10^{-3}$ at $r = 0.005$ per Math432, $\ll$ the diagonal margin away from the origin) is NOT recomputed at $r = 0.219$ here; a bracket continuum bound at $r = 0.219$ is required for an ANCHORED (exact) continuum closure. The present result is strong-evidence grade. The gate stays OPEN.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) “*The diagonal bound is not the exact ($\Delta F_{\text{anchored}}$) bound.*” UPHELD and named: the bracket is the off-diagonal correction; since it is negative, $\Delta F_{\text{anchored}} < \Delta F_{\text{diag}}$, so a diagonal bound does not by itself close the anchored no-condensate. The diagonal margin ($\gtrsim 10^{-4}$ away from the origin) exceeds the Math432 bracket scale (7.6×10^{-3} only at the largest amplitudes, where the diagonal value is ~ 0.03), so anchored positivity is expected, but the $r = 0.219$ bracket continuum bound is the named residual.

(β) “*Is the M -minimisation grid coarse enough to miss a dip?*” DISMISSED at strong-evidence grade: the 2D curvature-chord bound is node-free in (A_1, A_2) , and the M -grid is the variational trial parameter (finer M only LOWERS ΔF further toward the true minimum, already captured as $+3.9 \times 10^{-5} > 0$). A finer M -grid is a cheap follow-up.

(γ) “*Does the soft-direction correction threaten the selection?*” DISMISSED: $\kappa_{\{200\}} = 3.86 > 0$, so the $(0,0)$ Hessian remains positive-definite and Reading-H is still a strict two-shell minimum; only the which-shell-is-softer bookkeeping changes, not the PD conclusion or B1 T6.

Quantitative sanity check (mandatory): magnitude – $\kappa_{\{110\}} = 5.16$ matches the single-shell BCC 5.116 at the B1 point (offset $< 1\%$ = fixed- M vs gap-solved); sign-direction – the surface min $+3.9 \times 10^{-5}$ and continuum bound $+1.2 \times 10^{-4}$ are positive (no-condensate direction); limit-case – the evaluator reproduces Math432 at $r = 0.005$ to 1.6×10^{-7} and $\Delta F(0,0) = 0$; consistency – $r = 0.219$ is stiffer than $r = 0.005$, so the no-condensate margin is expected to be at least as large.

7. Result footer

Result ID: B1-RH-ENUM / two-shell global continuum no-condensate
at the B1 point (diagonal) + 2 corrections

Precise statement: At $r_{\text{bare}}=0.219$ (B1): the diagonal-continuum two-shell
{110}+{200} free energy, M-minimised, has min over
(A1,A2)!(0,0) = +3.9e-5 > 0, a 2D curvature-chord
continuum lower bound +1.2e-4 > 0 away from the origin,
and a positive-definite (0,0) Hessian ($\kappa_{\{110\}}=5.16$,
 $\kappa_{\{200\}}=3.86$). Evaluator validated vs Math432 ($r=$
0.005) to 1.6e-7. Corrects estimator-upgrade-knobs v1.0:
(1) Math432 evidence is at $r=0.005$ not the B1 point;
(2) soft eigenvalue = $\kappa_{\{200\}}=3.86$ not 5.116.

Scope: Two-shell {110}+{200}, DIAGONAL-continuum free energy,
B1 point. Exact-Wick bracket NOT included.

Dependencies: Math424_AddA_reading_uniqueness (M_fast, dl, gap_solve);
Math432 (validation anchors + bracket scale).

Evidence grade: EXECUTED (twoshell_continuum_bound.py 10/10) +
STRONG-EVIDENCE (diagonal-continuum bound).

Reproduction command: python codes/vacuum/twoshell_continuum_bound.py
10/10 PASS; min_off=+3.9e-5; continuum LB=+1.2e-4;
Expected output: $\kappa_{\{110\}}=5.16$, $\kappa_{\{200\}}=3.86$.

Falsification gate: A diagonal surface point with $dF<0$ at $r=0.219$; a negative
2D continuum lower bound; a non-PD (0,0) Hessian; or an
exact-Wick bracket at $r=0.219$ exceeding the diagonal
margin (would reopen the anchored no-condensate).

Tier before / after: No tier/gate flip. ESTIMATOR-UPGRADE: two-shell global
no-condensate ADVANCED (diagonal-continuum at the B1
point); gate stays OPEN pending the bracket continuum
bound + operator sign-off.

No-overclaim statement: DIAGONAL-continuum only; the exact-Wick off-diagonal
bracket at $r=0.219$ is the named residual. Strong-evidence,
not a theorem. The (0,0) Hessian PD conclusion of
estimator-upgrade-knobs v1.0 stands; its soft-eigenvalue
value and the cited operating point are corrected here.

Next required action: Exact-Wick bracket continuum bound at $r=0.219$ (the
slogdet engine), then ESTIMATOR-UPGRADE closure is an
operator decision.